

ASSIGNMENT - 6

- SAMPLE AND HOLD

Object

1. To demonstrate the operation of taking and holding samples of signals.
2. To study the effects of sampling time (sampling pulse width)
3. To study the effects of changing the sampling frequency.

Equipment required

- Sample and hold module 2960E
- Dual-beam oscilloscope, bandwidth 5MHz or better
- Audio-frequency signal generator
- Frequency counter
- Power supply : ED-2900P

Time required

One hour

Prerequisites

- Assignments - 1 and 2

INTRODUCTION

This assignment will introduce you to some of the basic ideas connected with sampling. The word 'sampling' can be used in many contexts, but for the present purpose the word is taken to refer to the representation of a continually varying analogue signal by a series of 'samples' or particular values of the signal measured at intervals in time. The idea is shown in Fig. 6-1, in which there are shown (a) a continuous line representing a signal which is to be sampled, and below it (b) a series of measurements of the signal at regular intervals Δt .

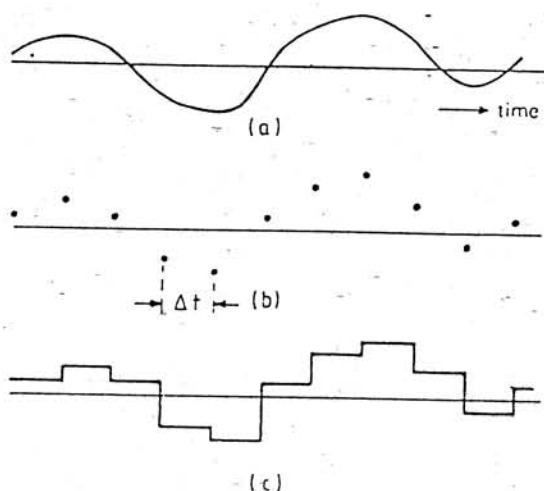


Fig. 6-1

Why should we want to sample a signal like this?

There are many reasons, depending usually on the means for transmitting or processing the signal, rather than on the signal itself. You will see in the next assignment how sampling is used to enable a communication channel to handle two signals at once. A radar often uses signals which are sampled from the beginning, since if a pulse is sent to a target and reflected, time must be allowed for the pulse to return before another pulse can be sent, and measurements of the target position can only be made at the same interval as the pulses are received. A computer, handling process data, may be occupied with arithmetic and other operations for a short time after receipt of a set of process measurements, and be unable during that time to receive the next set.

Whatever the reason for sampling, if samples are taken frequently enough, it would seem possible to reconstruct the original signal sufficiently well. But first, what is involved in taking each sample? Fig. 6-1 implies that

each sample is an instantaneous measurement, but in real life the taking of a measurement requires some time, even if short.

And a literally instantaneous sample might be of little use to a communication system, since it would contain no energy unless its amplitude were theoretically infinite. In practice it is necessary to stretch the sample out in time, or to hold on to it long enough to be able to use it. The Sample and Hold Module 2960E contains several pieces of equipment, not all of which will be used in this assignment. In particular it contains two sampling circuits. At the top left corner of the module's panel, input 1' is connected to an amplifier, whose output feeds a capacitor through a 'sampling switch'. The latter is an electronic switch which can be closed for periods of a few microseconds under the control of a signal, label 1' pulse 1'. If the analogue signal of Fig. 6-1 is varying as shown, and the switch is closed for a short time by applying a pulse to its control terminal, the capacitor will charge to the voltage of the analogue signal. At the end of the pulse, when the switch is opened once more, if there are no leakage paths for the charge on the capacitor to escape by, the capacitor will hold that voltage. The sampled voltage is in effect remembered by the capacitor. If a train of short pulses is applied to the switch, a series of different signal values will be sampled and held in turn, making a signal on the capacitor which changes in steps, as shown in Fig. 6-1(c).

In practice, when the switch is closed, the voltage on the capacitor cannot instantly take the value of the analogue signal. In order to change the capacitor voltage, a current must flow, whose value will be restricted by resistance in the switch and the signal source (in this case the output impedance of the amplifier). Thus time is necessary for the charge to change. How much time depends on the resistance R in the charging path, and the capacitance C which has to be charged.

To consider a very simple case, if during the time a sample is taken the analogue signal has a constant value k , and the stored previous value on the capacitor was j , then the capacitor voltage should ideally jump $(k - j)$, from j to k , but in practice, if the sample pulse width (time the switch is closed) is t , then the actual jump will be not $(k - j)$ but only

$$(k - j) (1 - e^{-t/CR}).$$

In effect the new signal value will not be fully taken in by the sample, and a fraction $e^{-t/CR}$ of the old sample will be left on the capacitor.

The signal is blurred: changes are made slowly, and, if the input signal changes before the output has changed fully, some signal is lost. To avoid this t/CR must be made sufficiently large. That is, the sample pulse width must be a few times greater than the charging time constant CR of the hold circuit. If the sample width is say $3 CR$, then only 5% of the jump is lost; if it is $4.6 CR$, only 1%.

The other factor this assignment will deal with is the frequency of sampling. It is easy to see that if enough samples are taken, the information in a signal can be contained in the samples. The question of the minimum number of samples required is obviously of interest, and the answer is less obvious. It turns out that :

When a signal is sampled at a rate such that at least two samples are taken during the period of its highest-frequency component, the samples can be used to reconstruct the original signal.

This principle is known as the Sampling Theorem.

To demonstrate the Sampling Theorem, we shall resort to a mathematical trick. It is one that previous assignments have prepared us for. Imagine that the signal is fed into a product modulator, or multiplier. If this modulator is to produce the same output as our sampler, the other input (the 'carrier') would be a series of narrow pulses, as in Fig. 6-2. This pulse train can be analysed by Fourier's Theorem into a harmonic series.

Call the waveform P , the pulse width dt (since it is supposed to be extremely narrow), and the pulse repetition frequency f .

Then $P = f_p dt (1 + \cos 2\pi f_p t + \cos 2.2\pi f_p t + \cos 3.3\pi f_p t + \dots)$

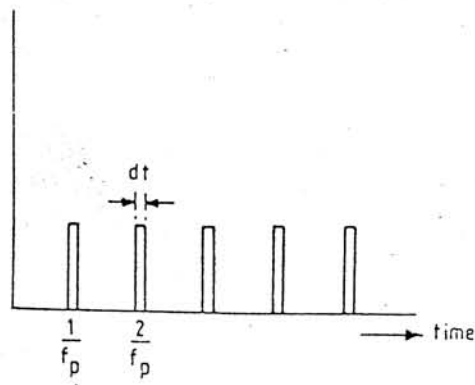


Fig. 6-2

The factor $f dt$ is a dimensionless multiplier and a constant. Each of the terms in the bracket will give rise to a term in the output of the modulator, and clearly the first term is the one which will cause the original signal to be present in its output, unchanged except for the multiplier $f dt$. Incidentally, it is because dt is supposed to be small to the point of vanishing that we have to stretch the pulses using the hold unit.

Since we have shown that the original signal is present in the modulator output (irrespective of the value of f_p), why the restriction about taking at least two samples? To answer that we must look at the other terms in the series P . Also, to simplify matters, let us take a simple signal $s = S \sin 2\pi f_s t$.

With this signal, the second term in the series P will give rise to a component in the output $s f_p dt \sin 2\pi f_s t \cos 2\pi f_p t$, which may be expressed as

$$s f_p dt \frac{1}{2} [\sin 2\pi(f_s + f_p)t + \sin 2\pi(f_s - f_p)t].$$

We have, so to speak, a pair of sidebands either side of the component frequency f_p of the 'carrier' P .

If we choose, f_s as the highest frequency we want to transmit, and f_p just equal to $2f_s$, then the lower 'sideband' becomes $s f_p dt \sin(-f_s)t$, which just cancels out the wanted signal. But worse follows if f_p is made any lower because the frequency of sideband also becomes lower, so that it represents a frequency lower than f_s , our highest wanted frequency. It is indistinguishable from another component of our input signal which could just as easily be present. This problem, and ways of tackling it, will be dealt with more thoroughly in the next assignment

PRACTICAL 6.1 - Effect of sampling pulse width

Plug the Module 2960E into the power supply and switch on the latter. Connect the 'internal clock output' on the module to the 'clock input' of its control logic.

Connect the frequency counter to the 'pulse 1' output of the control logic. In making this and all subsequent external connections, remember that a ground connection is necessary to complete the signal path.

Measure the frequency of the 'pulse 1' output. Transfer the frequency counter to the 'pulse 2' output and verify that its frequency is the same. It will be convenient to leave the frequency counter connected thus, to avoid several connections and disconnections. It will always correctly measure the frequency of the pulses at 'pulse 1' and 'pulse 2' outputs.

Adjust the 'clock frequency' control until the frequency counter indicates about 5kHz.

Connect the Y1 channel of the oscilloscope to the 'pulse 1' output, and adjust the 'sample pulse width' control to give a pulse width of about $5\mu\text{s}$. The width measurement should be taken at about mid-height of the pulse, which may be found to be rather rounded owing to a series resistor (which protects the Module circuits both from damage and from poor functioning because of external loading). Avoid holding the connection, which will make the rounding worse.

Remove the Y1 connection, and connect the Y1 channel, the audio signal generator, and the 'input 1' on the module all together. Set the signal generator to give a 100Hz sine-wave output of 3V peak-to-peak amplitude.

Connect the Y2 channel of the oscilloscope to the capacitor of the corresponding (upper) sample-and-hold circuit. Observe the waveform and compare it with the original. It may be necessary to make a slight readjustment to one of the frequencies to obtain a steady waveform, in which you should be able to see the sampling steps.

Turn the 'sample pulse width' control anticlockwise to decrease the sample pulse width.

Q 6.1 Why does the amplitude of the sampled output decrease?

Q 6.2 Why does the phase of the sampled output lag?

Rotate the 'sample pulse width' control clockwise to increase the width of the sample pulse.

Q 6.3 Why does the output not significantly increase above the first value you obtained (with the $5\mu\text{s}$ setting)?

Connect the capacitor output to the input of 'filter 1' on the module, and compare the waveform at the filter output with the original waveform.

Q 6.4 Does the filter generate an output waveform more like the original signal than its input, and if so, how?

Set the sample pulse rate to about 20kHz. Set the audio signal generator to give a 500Hz output at an amplitude of 5V peak-to-peak.

Set the pulse width to $25\mu\text{s}$. Measure the output voltage from the filter. You may find it convenient to do this using the other channel of the oscilloscope without altering the connections or time-base settings with which you obtained the pulse width setting.

Reduce the pulse width successively to 20, 15, 10, 5, 4, 3 and $2\mu\text{s}$. For each pulse width measure the peak-to-peak output voltage of the filter. Tabulate your results, and draw a graph of output voltage against pulse width.

Leaving the pulse width at $2\mu\text{s}$, change the sampling frequency over the whole available range, and note the maximum and minimum amplitudes of the filter output. Then increase the pulse width considerably, and repeat the observation.

Q 6.5 Can you explain the difference in behaviour in the two cases?

PRACTICAL 6.2 Relation between sampling and signal frequencies

Set the sampling frequency to 5kHz. Set the sampling pulse width to maximum (so that its exact value becomes unimportant).

Connect the oscilloscope Y1 channel to the audio signal generator, which should still be set to 500Hz. Set the oscilloscope to give a clear and steady display of this waveform.

Connect the Y2 channel to the filter output, and verify that a similar waveform is present.

Raise the frequency very slowly, and at some frequencies you will be able to see a superimposed higher frequency signal, which can be made to 'crawl' along the intended signal. At other frequency settings the superimposed spurious signal will become a blur, whose peak-to-peak amplitude can nevertheless be easily estimated.

Q 6.6 At what value of signal frequency does the spurious signal reach 10% of the desired signal?

Connect the output of Filter 1 to the input of Filter 2, the output of Filter 2 to the input of Filter 3, and so on, so that the signal is passed through all four filters in tandem. Observe the output of Filter 4.

Q 6.7 At what value of signal frequency does the spurious signal now reach 10% of the desired signal? How many times the sampling frequency is this signal frequency?

Continue raising the frequency of the signal generator, and observe that the output signal ceases to bear any resemblance to the input. Switch off all the equipment.